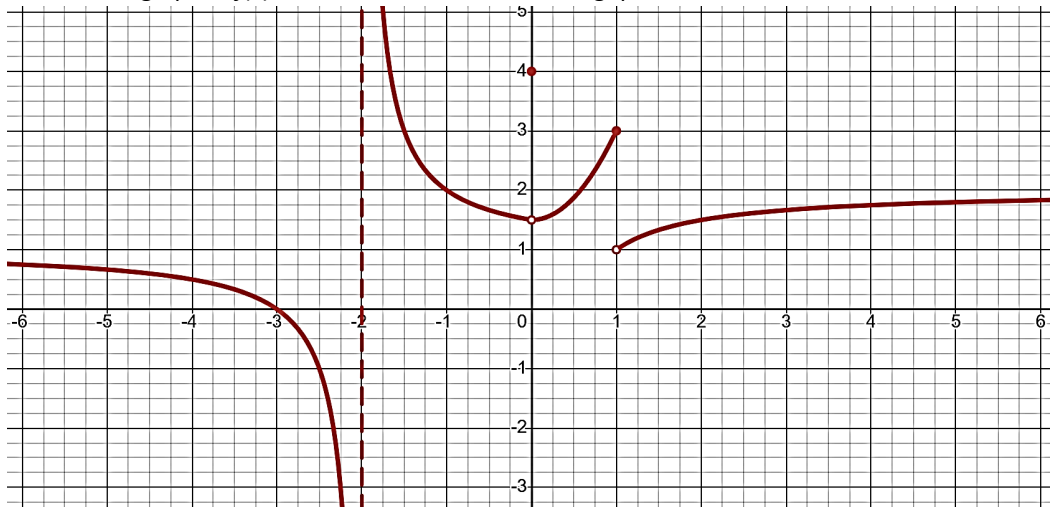


Show work if applicable. Use infinite limits to describe unbounded behavior.

1. Use the table below to evaluate the following limit: $\lim_{x \rightarrow 1} \frac{\sqrt{3x+1}-2}{x-1} =$ _____

x	0.9	0.99	1	1.01	1.1
$f(x)$	0.7646	0.7514	??	0.7486	0.7364

2. Use the graph of $f(x)$ below to answer the following questions.



a. $\lim_{x \rightarrow \infty} f(x) =$

b. $\lim_{x \rightarrow -2^+} f(x) =$

c. $\lim_{x \rightarrow -2^-} f(x) =$

b. $\lim_{x \rightarrow -2} f(x) =$

d. $\lim_{x \rightarrow 0} f(x) =$

f. $\lim_{x \rightarrow 1} f(x) =$

- g. Is $f(x)$ continuous at $x = 0$? Use the definition of continuity to explain why or why not.

3. Use $\lim_{x \rightarrow c} f(x) = 4$ and $\lim_{x \rightarrow c} g(x) = -3$ to find the following:

a. $\lim_{x \rightarrow c} [f(x) + 2g(x)] =$

b. $\lim_{x \rightarrow c} [f(x) \cdot g(x)] =$

c. $\lim_{x \rightarrow c} \left[\frac{g(x)}{f(x)} \right] =$

4. Find the x -values (if any) at which $f(x)$ is not continuous. Label them as removable or non-removable.

$$f(x) = \begin{cases} x^2 - 2, & x \leq 1 \\ \frac{x^2 - 4}{x - 2}, & x > 1 \end{cases}$$

5. Find the limit algebraically (if it exists).

a. $\sec \sec (2x)$

b. $\frac{\frac{1}{x+3} - \frac{1}{3}}{x}$

c. $\frac{3-x}{x}$

d. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{\sqrt{x+1} - 2} =$

e. $\lim_{x \rightarrow \infty} \frac{4x^2 - 1}{2x^2 + 5x}$

f. $\frac{x^2 - 5x - 14}{x - 7}$

g. $\frac{-3}{x^2 - 25}$

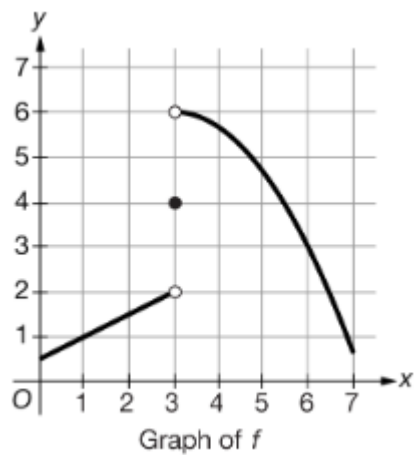
h. $\frac{5x+x}{x}$

6. Find a and b so that the function is continuous for all real values of x .

$$f(x) = \begin{cases} 3x + a & \text{if } x < -1 \\ ax - 2b & \text{if } -1 \leq x \leq 2 \\ b + 4x & \text{if } x > 2 \end{cases}$$

7. Verify that the Intermediate Value Theorem applies. If it does, use it to show that $f(x) = x^3 + x - 2$ has a zero on the interval $[0, 2]$. Then find the value(s) of c guaranteed by the theorem.

8. The graph of a piecewise function f is shown below. Which of the following expressions equals 2?



- (A) $f(3)$
 (B) $\lim_{x \rightarrow 3^-} f(x)$
 (C) $\lim_{x \rightarrow 3^+} f(x)$
 (D) $\lim_{x \rightarrow 3} f(x)$

9. For $x < 0$, the line $y = 3$ is an asymptote of the graph of a function f . Which of the following statements must be true?
- (A) $f(x) \neq 3$ for $x < 0$
- (B) $\lim_{x \rightarrow -\infty} f(x) = 3$
- (C) $\lim_{x \rightarrow \infty} f(x) = 3$
- (D) $\lim_{x \rightarrow 3} f(x) = \infty$
10. If a function f is continuous on the closed interval $[2, 5]$ and if $f(2) = -3$ and $f(5) = 4$, then which of the following must be true?
- (A) $f(c) = 2$ for some number c in the open interval $(-3, 4)$
- (B) $f(c) \neq 1$ for any number c in the open interval $(2, 5)$
- (C) $f(c) = 0$ for some number c in the open interval $(-3, 4)$
- (D) $f(c) = 2$ for some number c in the open interval $(2, 5)$
11. A student attempted to confirm that the function $f(x) = \frac{x^2 - 5x + 4}{x^2 - 6x + 8}$ is continuous at $x = 4$. In which step, if any, does an error first appear?
- Step 1: $f(x) = \frac{x^2 - 5x + 4}{x^2 - 6x + 8} = \frac{(x-4)(x-1)}{(x-4)(x-2)}$
- Step 2: $f(x) = \frac{x-1}{x-2} = \frac{3}{2}$
- Step 3: $f(4) = \frac{4-1}{4-2} = \frac{3}{2}$
- Step 4: $f(x) = f(4)$ so f is continuous at $x = 4$
- (A) Step 2
- (B) Step 3
- (C) Step 4
- (D) There is no error in the confirmation.